Information Security

SPRING 2023

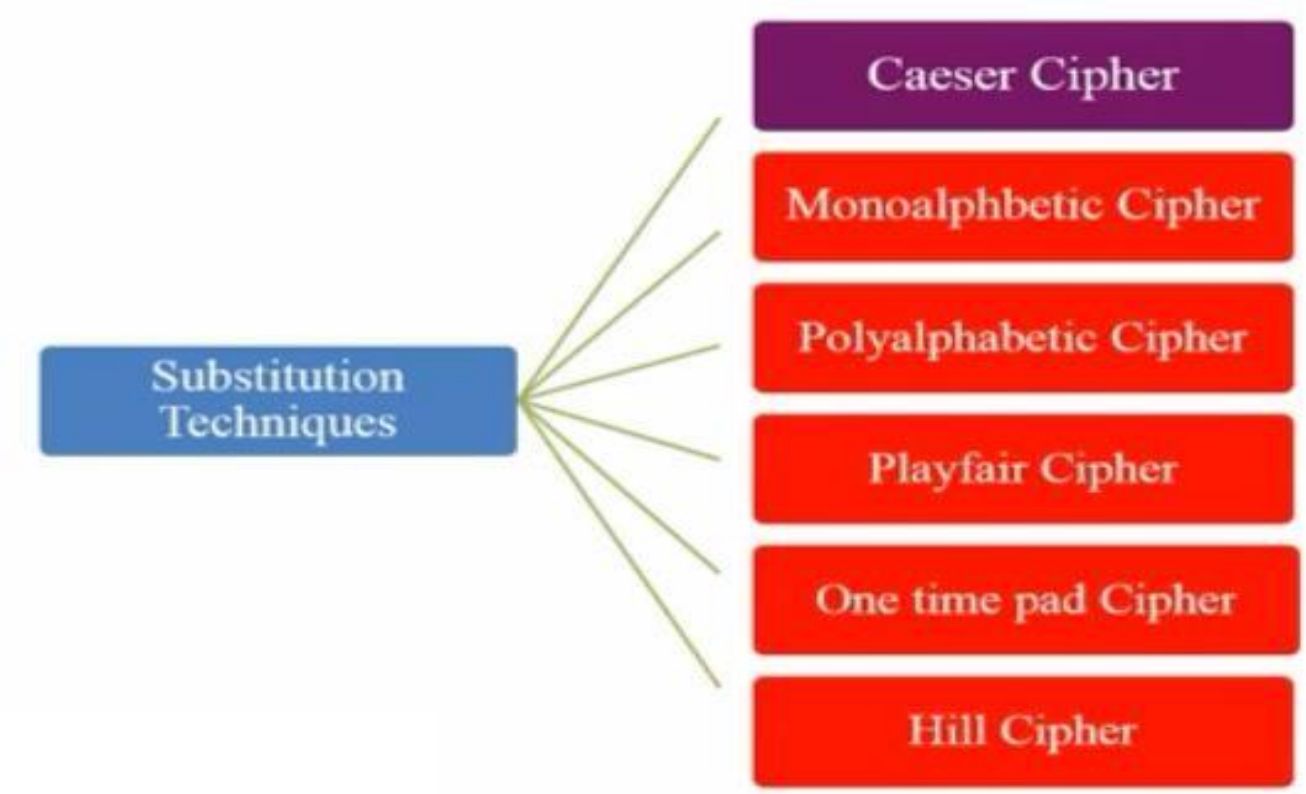
Lecture 6: Ceaser and Monoalphabetic Cipher

Introduction

- There are basically two types of symmetric cipher:
  - Substitution Cipher
  - Transposition Cipher
- **Substitution Cipher:** A substitution is a technique in which each letter or bit of the plaintext is substituted or replaced by some other letter, number or symbol to produce cipher text.
- ***For Example,*** ABC $\Longrightarrow$ XYZ
- **Transposition Cipher:** In transposition technique, there is no replacement of alphabets or numbers occurs instead their positions are changed or reordering of position of plain text is done to produce cipher text.
- ***For Example***, ABCDE $\Longrightarrow$ BADEC

Substitution Cipher

- Types of substitution techniques:



Ceaser Cipher

- Caesar cipher is type of substitution cipher. This technique was found by Julius Caesar. This technique is very simple and easy to generate cipher of given plain text. This technique is also known as shift cipher.
- In Caesar cipher techniques each letter/alphabet is replaced by the letter/alphabet which is three place next to the letter which is to be substituted. (Three place next to the letter method was used by Julius Caesar.)

Algorithm

- **Step-1:** The encryption can also be represented using modular arithmetic by first transforming the letters into numbers, according to the scheme, a = 0, b= 1, ..., z = 25. Encryption of a letter (C) by a shift (E)can be described mathematically.
- Generalize equation of Caesar cipher (Encryption & Decryption):
  - C = E (K, P) = (P + K) mod 26
  - P = D (K, C) = (C – K) mod 26
- Where, C = Ciphertext, P = Plain Text, E = Encryption, D = Decryption, K = Key

Algorithm

- **Step-2:** The transformation can be represented by aligning two alphabets; the cipher alphabet is the plain alphabet shift left or right by some number of positions. For instance, here is a Caesar cipher using a shifted of three places. (Here key is 3).

Key = **3**

Plain text = **"C"** → Cipher text= **"F"**

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12
n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

Algorithm

- **Step-3:** When decrypting, a person looks up each letter of the cipher text in the and writes down the corresponding letter in the “plaintext" line (Key = 4). Deciphering is done in reverse, with a right shift of 4.

Example, Cipher text = “h” , *Key* = *4*

plain text = “d”

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12
n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

Example - 1

- **Question:** Find out encrypted test using ceaser cipher.
  - Plain text: man, key: 3
  - Solution:

Plain text:- “m” ➔ Cipher text:- “p”

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12
n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

Example - 1

Plain text:- "a" ➔ Cipher text:- "d" **(Key = 3)**

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12
n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

Plain text:- "n" ➔ Cipher text:- "q" **(Key = 3)**

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12
n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

Cipher Text: pdq

Class Task

- Convert plain text into cipher text
  - Plain Text: meet me after part
  - Key: 5
  - Cipher Text: ???

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12
n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

Monoalphabetic Cipher

- A *monoalphabetic substitution cipher,* also known as a simple substitution cipher, relies on a fixed replacement structure.
- That is, the substitution is fixed for each letter of the alphabet.
- Thus, if “a” is encrypted to “R”, then every time we see the letter “a” in the plaintext, we replace it with the letter “R” in the ciphertext.
- Ceaser cipher is also monoalphabetic cipher.

Monoalphabetic Cipher

- A simple example is where each letter is encrypted as the next letter in the alphabet:
- For example,
- Plain Text: hidden treasure
- Cipher Text: IJEEFO USFBTVSF

a	b	c	d	e	f	g	h	i	j	k	l	m
B	C	D	E	F	G	H	I	J	K	L	M	N
n	o	p	q	r	s	t	u	v	w	x	y	z
O	P	Q	R	S	T	U	V	W	X	Y	Z	A

- The table below shows how one might choose to, and we will, lay them out for this example.

Class Task

- Convert plain text into cipher text by considering two tables which given below:
- Perform two rounds
- Follow table 1 for first round and table 2 for second round
- Plain Text: meet me after party

a	b	c	d	e	f	g	h	i	j	k	l	m
B	C	D	E	F	G	H	I	J	K	L	M	N
n	o	p	q	r	s	t	u	v	w	x	y	z
O	P	Q	R	S	T	U	V	W	X	Y	Z	A

A	B	C	D	E	F	G	H	I	J	K	L	M
d	e	f	g	h	i	j	k	l	m	n	o	p
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
q	r	s	t	u	v	w	x	y	z	a	b	c

Summary of today's lecture

- We discussed two types of symmetric cipher substitution and transposition cipher.
- We also discussed two types of substitution cipher ceaser cipher and monoalphabetic cipher.